Reg. No. | | | | | | | | | | | [18MFT3001]

THIRD SEMESTER UNIVERSITY EXAMINATION, DECEMBER - 2021

SCIENCE OF BEVERAGES

Time:03 Hours **Maximum Marks:70**

Instructions: Answer all questions. For questions 1 through 4, answer either parts (a) and (b) OR parts (c) and (d)

1. a. Compare the relationship between water activity and shelf-life of food. 06
   b. Discuss the water vapour sorption isotherm. Explaining different methods for determination of sorption isotherm. 08

   OR

   c. Discuss the role of a_w in microbial growth among food samples. 06
   d Describe different properties of water and their impact on water characteristics. 08

2. a. Differentiate different water softening processing along with their advantages and Disadvantages. 08
   b. Classify different types of impurities present in source water. 06

   OR

   c. With the help of appropriate examples, discuss the impact of iron and manganese contamination in water and their treatment process. 08
   d. What are the key difference between the sedimentation process and Ion Exchange process. 06

3. a. Distinguish between different types of fruit based beverages. 08
   b. Examine different component of the beverage. 08

   OR

   c. Outline different methods used for fruit juice concentration. 06
   d. " FSSAI and drinking water standards" critically comment on the statement 08

4. a. With help of flow chart, explain the chocolate production process 08
   b. Write critical comments on (3+3)
      i. Chemical changes during beer production
      ii. Chemistry of wine

   OR

   c. With the help of a flow chart, distinguish between dry and wet processing of coffee. 08
   d. Write critical comments on (3+3)
      i. Difference between green tea and black tea
      ii. Distilled alcoholic beverages

5. **Write short notes on any FOUR of the following.** 4 X 3.5=14
   a. Acidulent
   b. Sweetners
   c. Preservatives
   d. Falvouring compounds
   e. Food Colour
   f. Water as constituent in soft drinks

Deemed-to-be University Learning Resource Centre Jain Global Campus